

QINHEPING HU

Palo Alto, CA ♦ (608) · 338 · 7031 ♦ hqheping@gmail.com
<http://pages.cs.wisc.edu/~qhu28/>

EMPLOYMENT

Axiom Math 2026-present

Member of Technical Staff

- Building a frontier large language model for mathematical problem-solving and software verification.

Bytedance Inc. 2025-2026

Software Engineer

- Worked on automatically generate unit tests using large language model. Knowledge in the theory of programming language, generative AI and prompt engineering is required to perform the job.

Amazon Inc. 2021-2025

Applied Scientist II @ Automated Reasoning Group (ARG)

- Developed loop-invariants support in the program verification tools CBMC (C bounded model checker) and Kani (verification for rust). Knowledge in the theory of programming language, formal method, and program verification is required to perform the job.
- Developed a program synthesizer to automatically generate loop invariants for Rust and C programs. Knowledge in program synthesis is required to perform the job.
- Worked on verification of the safety of the Rust standard library, which is the most popular library in Rust.
- Worked on introducing automated reasoning checks to the Large Language Model (LLM), which can help users generate a more accurate LLM response on highly regulated topics. Knowledge in generative AI and prompt engineering is required to perform the job.

EDUCATION

University of Wisconsin-Madison, Madison 2015-2021

Ph.D. in Computer Science, advised by Loris D'Antoni

Minor in Mathematics: Probability, Statistics, Stochastic Processes, and Category Theory

- My Ph.D. study is about extending a kind of popular synthesis problems (syntax-guided synthesis) with quantitative objectives and presenting an algorithm to solve such extended problems.

Tsinghua University, Beijing, China 2011-2015

B.E. in Computer Science, Special Pilot Class (Yao Class), advised by Prof. Giulio Chiribella

PUBLICATION

Kani: A Model Checker for Rust ASE 26

Remi Delmas, Zyad Hassan, Qinheping Hu, Rahul Kumar, Felipe R. Monteiro, Thanh Nguyen, Adrian Palacios, Celina Val, Michael Tautschnig, Justus Adam, Daniel Schwartz-Narbonne, Carolyn Zech

Programmable Program Synthesis *Invited keynote* CAV 21

Jinwoo Kim, Qinheping Hu, Loris D'Antoni, Thomas Reps

Synthesis with Asymptotic Resource Bounds CAV 21

Qinheping Hu, John Cyphert, Loris D'Antoni, Thomas Reps

Semantics-guided Synthesis Jinwoo Kim, Qinheping Hu, Loris D'Antoni, Thomas Reps	POPL 21
Exact and Approximate Methods for Proving Unrealizability of Syntax-Guided Synthesis Problems Qinheping Hu, John Cyphert, Loris D'Antoni, Thomas Reps	PLDI 20
Solving Program Sketches with Large Integer Values Rong Pan, Qinheping Hu, Rishabh Singh, Loris D'Antoni <i>Selected for special issue of TOPLAS</i> <i>Nominated for EAPLS Award for the best ETAPS paper on PL and systems</i>	ESOP 20
Automatic Repair of Regular Expressions Rong Pan, Qinheping Hu, Gaowei Xu, Loris D'Antoni	OOPSLA 19
Direct Manipulation for Imperative Programs Qinheping Hu, Roopsha Samanta, Rishabh Singh, Loris D'Antoni	SAS 19
Proving Unrealizability for Syntax-Guided Synthesis Qinheping Hu, Jason Breck, John Cyphert, Loris D'Antoni, Thomas Reps	CAV 19
Syntax-Guided Synthesis with Quantitative Syntactic Objectives Qinheping Hu, Loris D'Antoni	CAV 18
Automatic Program Inversion using Symbolic Transducers Qinheping Hu, Loris D'Antoni	PLDI 17
Units of rotational information Yuxiang Yang, Giulio Chiribella, Qinheping Hu	New J. Phys. 17

WORKSHOP

Guarantees in Program Synthesis Qinheping Hu, Jason Breck, John Cyphert, Loris D'Antoni, Thomas Reps	SYNT 19
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RESEARCH EXPERIENCE

Research Intern, Microsoft Research Summer 2017

- Designed and developed a framework to utilize NLP in fault localization and program repair. We took a subset of Microsoft Driver as our target, mutated them to produce more correct/buggy programs. And we used our model to decide whether a given program is buggy, where are the bugs, and what are possible fixes for the bugs.
- Mentor: Rishabh Singh.

Bachelor Thesis 2014-2015

- *Quantum Fission of Maximally Entangled Spins*
- This work is part of the paper Units of rotational information.

Semester Thesis Spring 2014

- *Survey on Partial Satisfaction* **Download**
- We survey the results of lower bounds of satisfied fraction in a k -satisfiable CNF formula.
- Advisors: Timon Hertli, Emo Welzl in ETH.

HONOR AND AWARD

Baidu Academic Scholarship	2011 - 2013
Tsinghua Scholarship for Special Pilot Class	2011 - 2015
Chinese High School Mathematics League · <i>first prize</i>	2010, 2011

SERVICE

Program Committee	CAV 2026, VMCAI 2024, SYNT 2021
External Reviewer	SAS 2026, FMCAD 2020
Artifact Evaluation Committee	PLDI 19

SKILLS

Language (Computer)	: C/C++ • Python • Rust • Shell
Language (Human)	: Chinese (native) • English (fluent)
Formal Method	: Program Verification • Program Synthesis • SMT Solver
Scientific Computing	: Numpy
Machine Learning	: Pytorch • statsmodels • pandas